

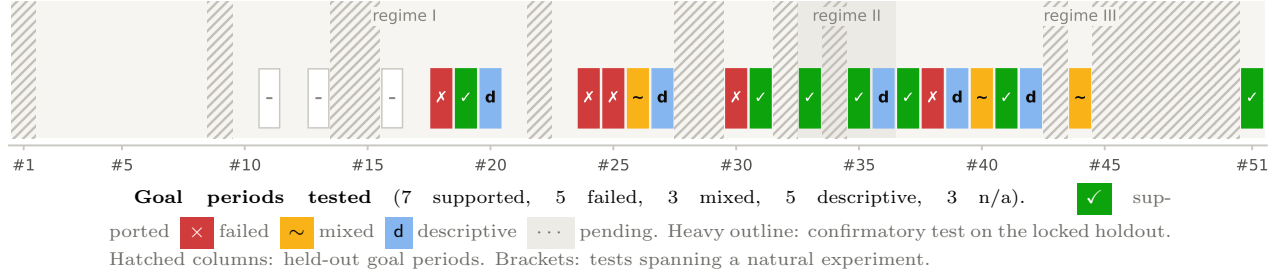
H11: Division of labor vs herding is the sign of a Potts coupling, set by the goal

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 60% Faithfulness 1.5/5 Usefulness 2.5/5

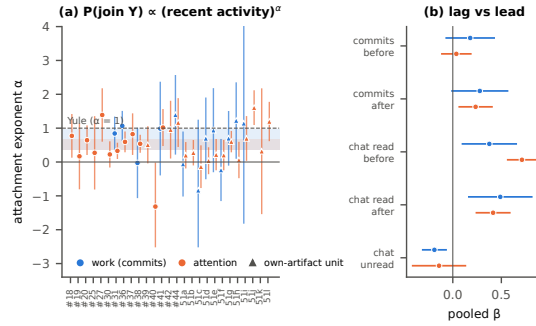
Bottom line. LLM agents pile onto projects in same-hour bursts steered by chat, not by the artifact record, and each extra agent on a shared project adds about 0.7 of a solo agent's commits.

level: hypothesis · pipeline checks 86% · updated 2026-10-05



WHAT HAPPENED

Rounds 1 and 1b fitted a Curie–Weiss Potts model per week. The spin is an agent's project in a 30-min window. The goal-mode rule for the sign of βJ failed: agents herd ($\beta J > 0$) in 11/14 weeks, in attention and in work commits, and spread only when the task gives each agent its own artifact. **Round 2 asks why agents pile on.** The degrees of freedom are *joins*: an agent switches to a project that others already hold. We fit the choice among all live projects with a conditional logit. Its attachment exponent α says how the odds grow with others' recent activity. Lead placebos test the arrow of time. Chat reads (context ledger) and commits (DQ4) compete as exposures, and a skeleton replay tests the project-size law. Output per agent comes from the work ledger. In 40 units, joins follow a sublinear kernel ($\alpha \approx 0.5$ – 0.7). Activity *after* a join predicts it as well as activity before, so this is co-arrival. Read chat beats commits, and a project's commits grow as $n^{0.70}$ with its agents.



Observables. (a) Attachment exponent α per unit with 95% CI, in work joins (blue) and attention joins (orange). Bands: pooled values, 0.68 [0.38, 0.98] and 0.52 [0.36, 0.67]. Both exclude the Yule value 1 (dashed). (b) Pooled exposure coefficients. Commits and chat reads *after* the join predict it as well as those before it. Read chat is the strongest predictor, and unread mentions have a negative coefficient.

PHYSICAL MODEL

Model 10 (Potts) in kinetic form, with a preferential-attachment kernel. Agent i joins project Y from the live set C . a_Y is other agents' labelled windows on Y in the last 2 active hours, and s_Y their cumulative windows. $h_{iY} = 1$ if i held Y before (habit). $\alpha = 1$ with $u \propto \log s_Y$ is the Yule–Simon process. The exposures are M (chat mentions of Y read at i 's calls in the last 60 min, or after the join: lead), M^u (posted but unread) and C (others' commits to Y , before or after). Output: c_Y is a project's commits in a window with n_Y agents on it, and $\theta < 1$ means crowding.

$$P(i \rightarrow Y) = e^{u_Y} / \sum_{Z \in C} e^{u_Z}$$

$$u_Y = \alpha \log a_Y + \beta_0 [a_Y=0] + \gamma \log(1+s_Y) + \eta h_{iY}$$

$$\text{R2: } u_Y = \beta_M \log(1+M_Y) + \beta_C \log(1+C_Y) \text{ (+lead, unread)}$$

$$\mathbb{E} c_Y \propto n_Y^\theta, \quad \text{RR}_{\text{herd}} = \text{rate}_{\text{herd}} / \text{rate}_{\text{solo}}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Herding is co-arrival steered by chat, not stigmergy. Agents pick projects in proportion to others' recent activity, with a sublinear exponent. That activity is as predictive one hour after the join as one hour before, so agents converge in the same hour; they do not follow an accumulated record. Read chat mentions predict joins better than commits, and unread mentions do not. Pile-ons cost little per herded agent (commit rate RR 1.23, n.s.), but they add little to the project: each extra agent adds about 0.7 of a solo agent's commits. Operator rule: expect a project's commits to grow as $n^{0.7}$ with the agents on it. To get parallel output, split work across owned artifacts rather than adding agents to one repo.

H11-potts-labor-vs-herding · generated 2026-10-04 20:49 by infra/summaries/build_summaries.py · card hypotheses/H11-potts-labor-vs-herding/README.md · ratings: H11 round-2 agent (suggested 60 / 1.5 / 2.5; previous calibrated 50 / 1.5 / 2.5)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 sign by goal mode (round 1)	5/11 signs; herding in 11/14 weeks	failed
R1a $\alpha > 0$ in $\geq 2/3$ shared (work)	2/4 work; 8/12 attention	failed (work)
R1b sublinear, Yule	not pooled α 0.68 / 0.52, both < 1 ; per-unit CIs wide	mixed
R1c fitness takes part	$\alpha_{FE} < \alpha$ in 5/13, 16/27	failed
R1d lag = lead (convergence)	CI $\ni 0$ in 34/40 units; attachment world gives +1.0 to +1.6	supported
R1e kernel replays sizes	work 10/13; attention 7/27	supported (work)
R1f Yule fits	2/13, 4/27	failed
R2a commits beat chat (held out)	work 6/13, pooled -0.002; attention 3/24, pooled -0.047 [-0.070, -0.024]	failed
R2b MH: commits > chat	pooled OR chat 3.0 vs commits 1.26 (attention)	failed
R2c commits lag > lead	work -0.18; shared work -0.78 [-1.22, -0.35]	failed
R3a herding raises output	RR 1.23 [0.94, 1.62] (shared)	not supported
R3b $\theta < 1$	16/23; θ 0.70 shared, 0.25 own	supported

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

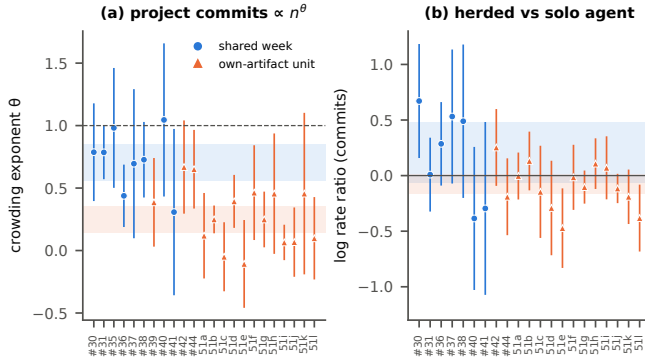
CAVEATS

Work joins are few (25–101 per period; 51g 321), so per-period work CIs are wide. Whole goal periods are units for #18–#44 (exception (d)). The 30-min window is coarser than the 18–50 s read-out delay, so “before” and “after” share a burst. The read vs unread contrast is not estimable in most units: unread mentions are almost never followed by a join. θ counts watchers as well as committers. Own units produce $3\times$ more commits per agent-hour, but that difference follows calendar time. The R3 test is mildly liberal (synthetic size 0.04–0.13). Synthetic validation found and fixed a likelihood bug and two liberal contrasts before the real run (amendment A1).

ROUND 2

Move the join test to the call clock. Compare commits visible at the deciding call with commits made during it (an artifact in-flight placebo). Recompute θ with committers only and file-level overlap, to split watchers from duplicated work. Use #40 ($\alpha < 0$: agents leave active projects) and NE42 as natives.

- **Essence**. Why do agents pile on: imitation, or attraction to where the work already is?
- **H11-R1** Preferential attachment: $P(\text{join } X) \propto (\text{recent activity on } X)^\alpha$.
- **H11-R2** Herding is stigmergic, not social: artifact activity predicts joining better than chat mentions of X (E3, vs H28).
- **H11-R3** Herding is a coordination solution that raises output; own-artifact spread duplicates effort (test against the work ledger).



Second figure. (a) Crowding exponent θ : a project’s commits per window vs the number of agents on it (project-day fixed effects). Shared weeks (blue) pool to 0.70 [0.56, 0.85], own-artifact units (orange) to 0.25 [0.14, 0.36]. In own units, attention to another agent’s repo is reading, not work. (b) Herded (≥ 2 others on the same project) vs solo windows of the same agent-day: log commit-rate ratio. Shared +0.21 [-0.06, 0.48]; own -0.08 [-0.17, 0.01].

GOAL PERIODS IN DETAIL

G18 $\times$ 1b: P1 failed (sig.); P2 failed (sig.) (unchanged)
G19 $\checkmark$ 1b: P1 supported; P2 supported (unchanged)
G20 d 1b: descriptive (unchanged)
G24 $\times$ 1b: P1 failed; P2 failed (sig.) (unchanged)
G25 $\times$ 1b: P1 failed (sig.); P2 failed (sig.) (unchanged)
G26 $\times$ 1b: mixed (native: jump on the right vote; cascade, n...
G27 d descriptive (first tested in round 2)
G30 $\times$ 1b: P1 failed (sig.); P2 failed (sig.) (unchanged)
G31 $\checkmark$ 1b: supported (native: work herds)
G33 $\checkmark$ 1b: supported (work herds)
+13 more periods: see the card.

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